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Increased volume of organ offers and decreased efficiency of kidney placement under circle-based kidney allocation

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Abstract

The newest kidney allocation policy ("KAS250") broadened geographic distribution while increasing allocation system complexity. We studied the volume of kidney offers received by transplant centers and efficiency of kidney placement since KAS250. We identified deceased-donor kidney offers (N=907,848; N=36,226 donors) to 185 U.S. transplant centers from 1/1/2019–12/31/2021 (policy implemented 3/15/2021). Each unique donor offered to a center was considered a single offer. We compared monthly volume of offers received by centers, and number of centers offered before first acceptance, using an interrupted time series approach (pre-/post-KAS250). Post-KAS250, transplant centers received more kidney offers (level change 32.5 offers/center/month, P<0.001; slope change 3.9 offers/center/month/month, P=0.003). Median monthly offer volume post-/pre-KAS250 was 195 (IQR 137–253) vs. 115 (76–151). There was no significant increase in deceased-donor transplant volume at the center level after KAS250, and center-specific changes in offer volume did not correlate with changes in transplant volume (r=−0.001). Post-KAS250, the number of centers to whom a kidney was offered before acceptance increased significantly (level change: 1.7 centers/donor, P<0.001; slope change: 0.1 centers/donor/month, P=0.014). These findings demonstrate the logistical burden of broader organ sharing, and future allocation policy changes will need to balance equity in transplant access with operational efficiency of the allocation system.

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INTRODUCTION

The shortage of kidneys available for transplant remains a challenge for the 89,000 prevalent patients waitlisted for a kidney transplant in the United States (US).[1] Given the scarcity of this life-saving resource—with 25,498 kidney transplants performed in 2022—allocation policy in the U.S. has been intentionally designed and periodically revised to improve equity while maximizing utility in the prioritization of patients for transplant. While allocation changes introduced in 2014 reduced racial disparities in transplant rates for waitlisted patients[2,3], large geographic variation in access to kidney transplant persisted.[4,5] These geographic disparities drew national and judicial system attention in 2017 when Miriam Holman brought a lawsuit against the Department of Health and Human Services for being unfairly deprioritized for lung transplant based on her place of residence.[6] Thereafter, in response to federal mandate, allocation policy was ultimately revised in March 2021 to enable broader sharing of deceased-donor kidneys in an effort to improve geographic disparities.[7,8]

Historically, donor kidneys were first allocated “locally” to waitlisted patients within the donation service area (DSA) where the kidney was recovered (Figure1A). However, there was significant variation in organ availability[5] and transplant access across DSAs.[9] The newest iteration of kidney allocation policy— “KAS250”—eliminated the geographic boundaries of the 57 DSAs from allocation prioritization, replacing them with 250 nautical mile circles centered on the donor hospital (Figure1B) while still allowing proximity points for transplant centers closer to the donor hospital. Yet, with this broader sharing comes greater operational complexity—in terms of workload for transplant centers[10] and logistical burden of allocation and organ distribution for organ procurement organizations (OPOs)[11]—as centers and OPOs learn to operate within new, broader networks.[12] These concerns are a threat to utility of the organ donation system, as an inefficient allocation system is not best equipped to serve our patients and may exacerbate the current problem of non-utilization of donor organs.[13] With upcoming and potentially more disruptive allocation changes being planned[14], there is an urgent need to evaluate KAS250’s implementation to identify potential inefficiencies in the system and inform the design of future iterations of allocation policy in the US.

We used national transplant registry data to study the impact of KAS250 on the efficiency of deceased-donor kidney allocation. We quantified trends in the number of kidney donors offered to transplant centers, hypothesizing that offers increased after KAS250. We then quantified the change in deceased-donor kidney transplant (DDKT) volume across transplant centers to evaluate the relationship between changes in offer vs. transplant volume. We also measured the efficiency with which a donor kidney is placed with an accepting center, hypothesizing that donors must be offered more widely under KAS250 and the resulting cold ischemia time would be longer.

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METHODS

Data source and study population

This study used data from the Scientific Registry of Transplant Recipients (SRTR). The SRTR data system includes data on all donor, wait-listed candidates, and transplant recipients in the U.S., submitted by the members of the Organ Procurement and Transplantation Network (OPTN). The Health Resources and Services Administration (HRSA), U.S. Department of Health and Human Services provides oversight to the activities of the OPTN and SRTR contractors. Candidate, recipient, and donor data from the standard analysis file were linked to the potential transplant recipient file, which includes all deceased-donor kidney offers to candidates, including the priority order and decision made for each offer (accept/decline). We included all deceased-donor offers made between 1/1/2019–12/31/2021 (most recently available data). We excluded centers receiving <10 offers in any given month and centers inactive in the pre-/post-KAS250 periods (eFigure1). This study was approved by the University of Texas at Austin institutional review board (#00002097).

Exposure: policy change

KAS250 was implemented on 3/15/2021. When quantifying trends over time, outcomes were analyzed from 1/1/2019–12/31/2021. When discretely comparing metrics in the pre- vs. post-KAS250 periods, to ensure balanced time periods, we compared pre-KAS250 (3/15/2020–12/31/2020) to post-KAS250 (3/15/2021–12/31/2021).

Outcomes: offer and transplant volume

Each donor goes through a “match run” in which it is offered to multiple candidates across different centers until each kidney is accepted for transplant. Organs offer decisions are made by the centers on behalf of the patients (eTable1 for further explanation). For our primary analysis, we measured the median number of deceased *donors* offered per center per month—meaning one donor offered to three candidates at a center would be counted as a single donor offer to that center. This approach prevents offer volumes from being influenced by a center’s waitlist size, more accurately capturing the true cognitive workload required for centers to determine acceptability of donor organ quality. We limited to instances where centers responded to the offer (even if kidney was ultimately accepted at a higher position on match run), and we excluded offers not actually seen by centers (“bypasses”).[15] This definition is reflective of actual workload from the center/surgeon’s perspective. For example, a center might receive and review an offer at sequence 4 on the match run, but it could ultimately be accepted at sequence 1 through 3 by another center. Similarly, kidneys that do not end up being accepted by any center still are reviewed by many centers. As a sensitivity analysis, we defined an “offer” multiple additional ways (1: only transplanted kidneys; 2: excluding offers where kidney was accepted at an earlier position on the match run; 3: combination of the two) to assess if our results were robust to these various definitions.

We also measured the median number of DDKT performed per center. To assess if changes in offer volume applied equally to lower vs. higher quality kidneys, we separately stratified

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by the kidney donor profile index (KDPI, a percentile score ranging from 0 [greatest longevity] to 100 [lowest longevity]; reference years 2019 and 2020).[16] We also compared donor characteristics pre/post-KAS250 using donor age, donation after brain death vs. donation after circulatory death, KDPI, and donor terminal serum creatinine.

Outcomes: efficiency of kidney placement

Next, we quantified the efficiency of kidney placement by a) calculating the median number of transplant centers considering a donor before its first kidney was accepted for transplant, and b) the number of patients (across all centers) to whom the organ was offered before it was accepted. These analyses excluded kidneys procured for transplant but not ultimately transplanted. An organ being declined by many centers or patients is reflective of either organ quality, or more likely, the efficiency of the allocation system.[17] We also quantified the proportion of kidneys allocated "out of sequence"—defined by the refusal codes 861 ("operational OPO"), 862 ("donor medical urgency"), or 863 ("offer not made due to expedited placement attempt")—in which OPOs exercise discretion to bypass the match run's prioritization to place a kidney in expedited fashion.[18] Finally, we reported national trends in the following: total national DDKT volume, median cold ischemia time (hours the kidney spent ex-vivo), proportion of kidneys with delayed graft function (defined as needing dialysis within 7 days post-transplant), and the proportion of recovered kidneys procured but not used (for any reason).

Statistical analysis

To estimate the effect of KAS250 on our measures of interest (donor offer volume, center-level DDKT volume, and efficiency of placement), we used an interrupted time series approach using ordinary least squares regression and Newey-West standard errors to handle autocorrelation and possible heteroskedasticity.[19] Autocorrelation was tested with a Cumby-Huizinga test.[20] This approach accounts for existing temporal trends and quantifies the *level change* (immediate change at KAS250's implementation) and the *slope change* after KAS250. Each model adjusted for the following covariates: month, median donor age and KDRI each month, and proportion of donation after cardiac death donors recovered each month. We then computed the median or frequency of each outcome metric, and of the donor characteristics, for post- vs. pre-KAS250 periods, comparing their statistical significance using a Wilcoxon rank-sum test or Fisher's exact test (independent samples) or Wilcoxon signed rank test (paired samples).

Next, we assessed the association of center-level changes in median offer volume, offer quality, and offer acceptance rates (using SRTR's definition[21]) with changes in DDKT volume (Pearson correlation). We computed the proportional change in each of these metrics by dividing the post-KAS250 by pre-KAS250 value. For these correlational analyses only, we removed outlier centers greater than 3 standard deviations above the mean (Figure3 footnotes).

Finally, for transplant centers experiencing proportional changes in offer volume below vs. above the median, we compared the following center characteristics: pre-KAS250 kidney transplant volume and characteristics of prevalent waitlist populations at KAS250

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implementation (waitlist size and factors determining allocation priority: waiting time, proportion of highly sensitized patients, and median estimated post-transplant survival of the waitlist cohort[22]).

RESULTS

We identified 907,848 deceased donor kidney offers for 36,226 unique donors to 185 U.S. transplant centers from 1/1/2019–12/31/2021 (eFigure1). When divided into pre-/post-KAS250 periods for discrete comparisons, the pre-KAS250 period (3/15/2020–12/31/2020) included 192,152 offers for 9,395 donors, and the post-KAS250 period (3/15/2021–12/31/2021) included 363,025 offers for 10,717 donors.

National trends in transplant volume, ischemia time, delayed graft function, and non-use

National monthly DDKT volume increased over time throughout the study period, with an immediate increase (level change: 207.9 DDKT/mo., P=0.003) followed by a downtrend after KAS250 (slope change −49.0 DDKT/mo./mo., P=0.001; eTable2, eFigure2). Total DDKTs performed was 13,580 pre- and 14,751 post-KAS250 (Table1). Compared to pre-KAS250, the post-KAS250 period saw increases in median cold ischemia time (19.3 vs. 17.1 hrs., P<0.001), incidence of delayed graft function (30.8% vs. 28.3%, P<0.001), and non-use proportion (24.4% vs. 20.7%, P<0.001) (Table1).

Characteristics of deceased-donor kidney offers

Characteristics of the offered deceased donors pre-/post-KAS250 are reported in eTable3. Post-KAS250, donors were older (median 43 vs. 41 yrs., P<0.001), more likely donation after circulatory death (31.7% vs. 26.5%, P<0.001) and slightly lower quality (median KDPI: 50 vs. 47, P<0.001).

Trends in deceased-donor kidney offer and transplant volume

Figure2A shows the median number of deceased-donor kidney offers/transplant center/month over time. Offer volume increased significantly after KAS250 (level change: 31.8 offers/center/mo., P<0.001; slope change: 4.1 offers/center/mo./mo., P=0.002; eTable2). In contrast, at the center-level, there was no immediate change in the median number of DDKT/center/month following KAS250 (level change: 1.6, P=0.058), but there was a small downtrend (slope change: −0.3 DDKT/center/mo./mo., P=0.003; eTable2). Compared to the pre-KAS250 period, median offer volume was 70% higher in the post-KAS250 period (195 vs. 115 offers/center/mo., P<0.001) and median DDKT volume was similar (6 vs. 6 DDKT/center/mo., P=0.017; Table1). Similar increases in offer volume post- vs. pre-KAS250 were observed across ranges of kidney quality (Table1, interrupted time series results in eTable2). When alternate definitions of an organ offer were used as a sensitivity analysis, the overall results were similar, with a 70–90% proportional increase in offer volume post vs. pre-KAS250 and a significant level change (increase) in offer volume post-KAS250 (eTable4 and eFigure3A–D).

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Efficiency of kidney placement

Figure2B shows the trend in median number of centers to whom a donor must be offered prior to receiving the first acceptance. This number increased significantly after KAS250 (level change: 1.7 centers/donor, P<0.001; slope change: 0.1 centers/donor/mo., P=0.010; eTable2). Compared to pre-KAS250, kidneys were offered to twice as many centers before finding an accepting center in the post-KAS250 period (4 vs. 2, P<0.001), and similar increases were seen across KDPI strata (Table1). This metric varied across OPOs, as the median number of centers offered prior to first acceptance ranged from 1–6 pre-KAS250 and 1–12 post-KAS250.

The median sequence number—or number of candidates to whom a donor was offered prior to the first kidney acceptance—also increased significantly after KAS250 (level change: 2.5 positions, P<0.001; slope change: 0.2 positions/mo., P=0.001; eTable2; median 7 vs. 3, P<0.001; Table1). Similar increases were seen across KDPI strata (Table1). This metric varied across OPOs, as the sequence number at first acceptance ranged from 2–14 pre-KAS250 and 2–19 post-KAS250. Nationally, out of sequence kidney allocations have become more common. The proportion of all donors with an out of sequence allocation increased from 3.3% to 5.6% after KAS250 (P<0.001; interrupted time series analysis: level change 2.0 percentage points, P<0.001), and this practice is increasingly common since KAS250 (eFigure4).

Relationship between changes in center offer volume and transplant volume

The proportional change in offer volume following KAS250 ranged from 0.7 to 8.4 across centers (median 1.7, IQR 1.4–2.0; eFigure5), and change in DDKT volume ranged from 0.2 to 6.0 (median 1.0, IQR 0.8–1.5). At the center-level, change in offer volume was not correlated with change in DDKT volume (r=−0.001, P=0.988, Figure3A). The change in quality of organs offered to a center was also not correlated with change in DDKT volume (r=−0.0001, P=0.999, Figure3B). Centers who received more offers after KAS250 had an associated decrease in their offer acceptance rate (r=−0.366, P<0.001), but centers who increased their acceptance rate had an associated increase in DDKT volume post-KAS250 (r=0.532, P<0.001, Figure3C). Nationally, there was no statistically significant change in the covariate-adjusted offer acceptance rate over time or following KAS250 (eFigure6).

Characteristics of centers with smaller vs. larger changes in offer volume

Centers with the largest proportional increases in offer volume after KAS250 were predominantly located in areas densely populated with donor hospitals and transplant centers —namely the northeast and Midwest (Figure4A)—whereas centers with large increases in DDKT volume were more geographically dispersed (Figure4B). The median proportional change in offer volume observed by transplant centers varied 4.6-fold across DSAs (0.7 to 3.3). Table2 compares characteristics of transplant centers based on the degree to which their offer volume changed after KAS250. Centers who saw *greater* increases in *offer* volume after KAS250, compared to centers who saw *lesser* increases in offer volume, were lower *DDKT* volume centers at baseline (median pre KAS250 DDKT volume 49 vs. 73 annual DDKT per center, P=0.026), but were otherwise similar with respect to their waitlist population's characteristics (Table2).

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DISCUSSION

KAS250 has resulted in large increases in the number of kidney donor offers received by transplant centers. Although the total number of transplants performed nationally continues increasing over time—a welcome finding resulting from national efforts to procure more organs and increase transplantation—at the transplant center level, receiving more offers was not associated with an increased in DDKTs. Increased offer volume increases workload for transplant centers and staff[10] (thus distracting from other patient care activities), and an inelastic response between offers and transplants reflects the greater logistical burden of allocation for centers and OPOs. However, centers who responded by increasing their offer acceptance rate after KAS250 *did* experience an associated increase in DDKT volume. As further evidence of inefficiency while the allocation system adapts to this new policy, deceased-donor kidneys are now offered to twice as many transplant centers before an accepting center is found. In turn, cold ischemia time, proportion of transplanted kidneys with delayed graft function, and proportion of kidneys recovered but not used, have all increased since KAS250. These consequences of the recent kidney allocation change demonstrate the logistical burden of broader organ sharing and should be considered when designing the next round of changes planned under continuous organ distribution.

The increased complexity accompanying broader sharing was identified as a potential concern given the dramatic increase in the number of centers now considered “local”—i.e. given distance-based allocation priority in the first round of the matching algorithm.[12] Single-center reports describe a near doubling of offer volumes[10]—which we show is a shared experience nationally—and the time spent by transplant teams evaluating these offers has nearly doubled.[10] Even using conservative estimates of 9 minutes per offer, at one institution this increase resulted in 25 hours/month spent by physicians on this singular aspect of the allocation system (evaluating offers) and 91 hours spent by transplant coordinators for related tasks.[10] For the centers facing this increased offer-related workload, when it is not coupled with an increase in transplants it instead can distract from other patient care activities and divert effort that could be directed elsewhere to improve access to transplantation. Others have raised concerns over the efficiency of allocation under KAS250 and the impact on kidney utilization and outcomes.[11,13] Our study has identified the contribution of offer volume as a driver of inefficiency in this system, and we have quantified the extent of this problem nationally, which affects nearly all transplant centers.

The rate of non-use of deceased-donor kidneys has reached an all-time high despite calls to action from a White House Executive Order[23], a report from the National Academies of Sciences, Engineering, and Medicine[24], and a Center for Medicare and Medicaid Services collaborative.[25] Similar political imperatives[26] have stimulated OPOs to procure more donors—as reflected in the annually increasing national DDKT volumes—and coupled with the pursuit of more marginal quality donors, this likely explains the increasing trend in offer volume before KAS250. Yet, with longer cold ischemia times[27], and centers and OPOs overwhelmed by offer volumes, kidney utilization is worsening.[13], Beyond the allocation changes, OPOs and transplant centers also share a role and may not have been adequately prepared to respond to these changes. OPOs vary in their efficiency of organ placement, which may reflect differences in organ quality or network size, or differences in OPO

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practices around organ offers and placement. OPOs are increasingly resorting to out of sequence allocation to get kidneys placed in the face of longer match runs, greater distances, and longer cold ischemia time. Transplant centers, too, have differed in their preparedness for the allocation changes. Preparation for a sudden increase in offer volume would include optimizing waitlist management (to ensure patients are ready for transplant), increasing staffing if possible, instituting offer filters, and possibly reconsidering one's acceptance criteria. Offer filters—which are underutilized across centers—can be prevent centers from being notified about kidneys they are unwilling to use. Offer acceptance rates of transplant centers are scrutinized and are now a performance metric. On the national scale, offer acceptance rates have not changed since KAS250, but some centers did increase their offer acceptance over this time, and these centers tended to increase their DDKT volume as well. In other words, centers who changed their behavior in response to the KAS250 tended to do more transplants over this time as well. Thus, we suggest that policies affecting distribution of kidneys must be coupled with changes in behavior and performance at the center and OPO level to improve kidney utilization.

The impetus for KAS250 was a need to improve geographic equity, which we have not measured with the present study. More kidney offers for transplant candidates is a positive aspect of allocation policy change *if* this translates to an increased likelihood of transplant for the patients most in need without overburdening the system or compromising kidney utilization. We show that simply receiving more kidney offers is not sufficient to increase the number of transplants at the center level. However, increasing transplant volume does not equate to increasing equity, and once sufficient post-KAS250 follow-up time has accrued, it will be critical to study the effect of broader distribution on geographic disparities in access to kidney transplant. Early results from the OPTN's one-year monitoring report[28] showed a 16% increase in the overall transplant rate following KAS250, with greater increases in certain marginalized sub-populations, but geographic disparities have not yet been assessed. These results are encouraging, and time will tell if the beneficial aspects of broader distribution outweigh the downsides highlighted in this manuscript. Ultimately, organ allocation seeks to balance *equity* in patient access to lifesaving donor organs with *utility* in maximizing their benefit (while also attempting to minimize non-utilization). Inefficiency and workforce strain in the transplant system can impede progress toward both goals; a focus on efficiency and practicality of allocation and organ distribution is necessary to align these goals and to better serve our patients.

KAS250 is the first iteration of broader geographic allocation, and there is room to refine this policy moving forward. Homogenous distribution circles ignore population size, disease burden, and organ availability; thus, a one-size-fits-all approach to distribution may be inadequate.[29] When we examined characteristics of the centers most affected by increased offer volume, these centers did not have differences in waitlist size or allocation priority of their candidates. Rather, geography was a major driver of the variation in extent of increased offer volume, and the largest increases in offer volume were seen in areas densely populated with donor hospitals and transplant centers. Further, the centers with greater increases in offer volume were smaller transplant centers who may not have the resources to adapt to these changes in offer volume, thus raising the potential for perpetuating or creating disparities in transplant access by favoring consolidation to the larger more well-resourced

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centers.[30] Many have concerns that kidneys are distributed away from areas of high need[31], including rural centers whose patients already have reduced access to transplant.[32,33]

KAS250 is an intermediate policy change *en route* to broader allocation changes termed "continuous distribution[8,14]," which builds on KAS250 by completely eliminating geographic boundaries and other fixed strata (like age group or allosensitization status) for a continuous scoring system. Continuous distribution is a more complex policy and will still share the broader distribution component of KAS250, but proximity score as a component of the scoring system will allow for titration of this parameter and thus the potential to respond to distance-based inefficiencies in the system. Nonetheless, important lessons can be learned from KAS250's implementation to inform these future iterations of allocation policy.

This study has several limitations. The timeframe of available data post-KAS250 was somewhat short; nevertheless, immediate and clear changes in the study outcomes and in their trends were observed and quantified. Whether the system will eventually adapt and see offer volumes decline or efficiency improve upon reaching more of an equilibrium[34] is unknown, and ongoing surveillance is needed. Second, the confounding impact of the COVID-19 pandemic cannot be discounted. The pre-KAS250 era coincided with the onset of the COVID-19 pandemic, and pre-/post-KAS 250 comparisons must be interpreted accordingly. The initial effect of the pandemic was a shutdown of transplant activity at many centers during the early months of the pandemic. Despite this, graphical display of pre-KAS250 trends revealed minimal effect on the national scale of COVID-19 onset on the study metrics. However, downstream effects of the pandemic on workforce shortages, for example, may still be contributing to inefficiencies in the transplant system. Third, additional confounders may limit our ability to attribute causality to the KAS250 policy. The donor pool is expanding as OPOs face new performance metrics and more marginal quality donor organs are procured and offered. Our time series models account for pre-KAS250 trends, and we adjusted for donor pool characteristics in our models, but still these other contemporaneous changes in organ procurement and transplantation may confound our results. Lastly, there are many ways to define an "offer" with these data, and we opted for a more inclusive definition to capture all offers a transplant center would have received and spent time reviewing. However, our definition may include some offers which may have been screened by a coordinator without the center spending time to review. As such, our estimates of number of offers reviewed per center per month may be biased higher than the actual offer volume, but the takeaway findings are robust to the various possible definitions as shown in our sensitivity analysis.

In this study of US transplant centers, since the implementation of KAS250—the most recent kidney allocation change to broader geographic distribution of kidneys—the volume of kidney offers received by transplant centers increased significantly, and the allocation system became less efficient. There was an increase in kidney cold ischemia time and in the kidney non-use rate during this same time-period. The effect of KAS250 on geographic equity in access to transplant was not the focus of our study but is a key question yet to be answered. Kidney allocation seeks to balance equity and utility, and an inefficient system is a threat to utility. As allocation policy continues to evolve, and with more substantial allocation changes planned ahead, these potential consequences should be recognized, and

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future policy iterations should balance efficiency of the system with equity in transplant access.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

DISCLOSURES:

The data reported here have been supplied by the Hennepin Healthcare Research Institute (HHRI) as the contractor for the Scientific Registry of Transplant Recipients (SRTR). The interpretation and reporting of these data are the responsibility of the author(s) and in no way should be seen as an official policy of or interpretation by the SRTR or the U.S. Government. S. Mohan reports consultancy for Angion Biomedica, eGenesis, and HSAG; an advisory or lead-ership role for ASN Quality Committee (member), ETCLC (NationalFaculty Chair), Kidney International Reports (ISN; deputy editor), SRTR Review Committee (member), and UNOS data advisory committee (vicechair); and research funding from NIH (NIDDK, NIHMD and NIBIB) and the Kidney Transplant Collaborative. The remaining authors have no potential conflicts of interest to disclose.

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Abbreviations:

US	United States
KAS250	“kidney allocation system 250” (recent allocation policy change to broader distribution)
DSA	donor service area
DDKT	deceased-donor kidney transplant
IQR	interquartile range
KDPI	kidney donor profile index
SRTR	Scientific Registry of Transplant Recipients
OPO	organ procurement organization

REFERENCES

1. United States Renal Data System. 2020 USRDS Annual Data Report: Epidemiology of kidney disease in the United States. National Institutes of Health, National Institute of Diabetes and Digestive and Kidney Diseases, Bethesda, MD, 2020.
2. Melanson TA, Hockenberry JM, Plantinga L, et al. New Kidney Allocation System Associated With Increased Rates Of Transplants Among Black And Hispanic Patients. Health Aff (Millwood). 2017;36(6):1078–1085. [PubMed: 28583967]

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3. Zhang X, Melanson TA, Plantinga LC, et al. Racial/ethnic disparities in waitlisting for deceased donor kidney transplantation 1 year after implementation of the new national kidney allocation system. Am J Transplant. 2018;18(8):1936–1946. [PubMed: 29603644]
4. Zhou S, Massie AB, Luo X, et al. Geographic disparity in kidney transplantation under KAS. Am J Transplant. 2018;18(6):1415–1423. [PubMed: 29232040]
5. King KL, Husain SA, Mohan S. Geographic Variation in the Availability of Deceased Donor Kidneys per Wait-Listed Candidate in the United States. Kidney international reports. 2019;4(11):1630–1633. [PubMed: 31891004]
6. Glazier AK. The lung lawsuit: a case study in organ allocation policy and administrative law. J Health & Biomedical L. 2018;14:139.
7. Israni A, Wey A, Thompson B, et al. New Kidney and Pancreas Allocation Policy: Moving to a Circle as the First Unit of Allocation. J Am Soc Nephrol. 2021;32(7):1546–1550. [PubMed: 34140395]
8. Notice of implementation: removal of DSA and region from kidney and pancreas allocation. Available at: https://unos.org/news/implementation-notice-dsa-region-removal-kidney-pancreas-allocation/ (2021). Accessed September 27, 2022.
9. Stewart DE, Wilk AR, Toll AE, et al. Measuring and monitoring equity in access to deceased donor kidney transplantation. Am J Transplant. 2018;18(8):1924–1935. [PubMed: 29734498]
10. Reddy V, da Graca B, Martinez E, et al. Single-center analysis of organ offers and workload for liver and kidney allocation. Am J Transplant. 2022.
11. Wood NL, VanDerwerken DN, Segev DL, Gentry SE. Increased Logistical Burden in Circle-based Kidney Allocation. Transplantation. 2022;106(10):1885–1887. [PubMed: 36173652]
12. Adler JT, Husain SA, King KL, Mohan S. Greater complexity and monitoring of the new Kidney Allocation System: Implications and unintended consequences of concentric circle kidney allocation on network complexity. Am J Transplant. 2021;21(6):2007–2013. [PubMed: 33314637]
13. Puttarajappa CM, Hariharan S, Zhang X, et al. Early Effect of the Circular Model of Kidney Allocation in the United States. J Am Soc Nephrol. 2022.
14. Continuous distribution: creating a more fair and patient-focused system for organ allocation. Available at: https://optn.transplant.hrsa.gov/policies-bylaws/a-closer-look/continuous-distribution/ (2022). Accessed September 27, 2022.
15. King KL, Husain SA, Cohen DJ, Schold JD, Mohan S. The role of bypass filters in deceased donor kidney allocation in the United States. Am J Transplant. 2022;22(6):1593–1602. [PubMed: 35090080]
16. Kidney Donor Profile Index Calculator - OPTN. https://optn.transplant.hrsa.gov/data/allocation-calculators/kdpi-calculator/. Accessed October 29, 2022.
17. King KL, Chaudhry SG, Ratner LE, Cohen DJ, Husain SA, Mohan S. Declined Offers for Deceased Donor Kidneys Are Not an Independent Reflection of Organ Quality. Kidney360. 2021;2(11):1807–1818. [PubMed: 35372993]
18. King KL, Husain SA, Perotte A, Adler JT, Schold JD, Mohan S. Deceased donor kidneys allocated out of sequence by organ procurement organizations. Am J Transplant. 2022;22(5):1372–1381. [PubMed: 35000284]
19. Linden A Conducting interrupted time-series analysis for single-and multiple-group comparisons. The Stata Journal. 2015;15(2):480–500.
20. Cumby RE, Huizinga J. Testing the autocorrelation structure of disturbances in ordinary least squares and instrumental variables regressions. National Bureau of Economic Research Cambridge, Mass., USA; 1990.
21. Scientific Registry of Transplant Recipients. Technical methods for the program-specific reports. Available at: https://www.srtr.org/about-the-data/technical-methods-for-the-program-specific-reports#tableb11 (2022). Accessed November 17, 2022. .
22. Estimate Post-Transplant Survival Calculator - OPTN. https://optn.transplant.hrsa.gov/data/allocation-calculators/epts-calculator/. Accessed October 29, 2022.
23. Executive Office of the President: Advancing American Kidney Health. Available at: https://www.federalregister.gov/documents/2019/07/15/2019-15159/advancing-american-kidney-health. Accessed October 24, 2022.

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24. National Academies of Sciences, Engineering, and Medicine. A Fairer and More Equitable, Cost-Effective, and Transparent System of Donor Organ Procurement, Allocation, and Distribution. In: Hackmann M, English RA, Kizer KW, eds. Realizing the Promise of Equity in the Organ Transplantation System. Washington (DC): National Academies Press (US), Copyright 2022 by the National Academy of Sciences; 2022.
25. Centers for Medicare and Medicaid Services. ESRD Treatment Choices (ETC) Model. Available at: https://innovation.cms.gov/innovation-models/esrd-treatment-choices-model (2022). Accessed November 21, 2022.
26. Lynch RJ, Doby BL, Goldberg DS, Lee KJ, Cimeno A, Karp SJ. Procurement characteristics of high- and low-performing OPOs as seen in OPTN/SRTR data. Am J Transplant. 2022;22(2):455–463. [PubMed: 34510735]
27. Barah M, Kilambi V, Friedewald JJ, Mehrotra S. Implications of Accumulated Cold Time for US Kidney Transplantation Offer Acceptance. Clin J Am Soc Nephrol. 2022;17(9):1353–1362. [PubMed: 35868843]
28. Organ Procurement and Transplantation Network, Kidney Transplantation Committee. Eliminate Use of DSA and Region from Kidney Allocation One Year Post-Implementation Monitoring Report. Accessed April 18, 2023. https://optn.transplant.hrsa.gov/media/p2oc3ada/data_report_kidney_full_20220624_1.pdf.
29. Karami F, Kernodle AB, Ishaque T, Segev DL, Gentry SE. Allocating kidneys in optimized heterogeneous circles. Am J Transplant. 2021;21(3):1179–1185. [PubMed: 32808468]
30. Adler JT, Husain SA. More is better … until it is worse: Can organ placement processes scale to an increasingly complex system? Am J Transplant. 2022.
31. DuBay DA, Morinelli TA, Su Z, et al. Association of High Burden of End-stage Kidney Disease With Decreased Kidney Transplant Rates With the Updated US Kidney Allocation Policy. JAMA Surg. 2021;156(7):639–645. [PubMed: 34037705]
32. McPherson LJ, Barry V, Yackley J, et al. Distance to Kidney Transplant Center and Access to Early Steps in the Kidney Transplantation Process in the Southeastern United States. Clin J Am Soc Nephrol. 2020;15(4):539–549. [PubMed: 32209583]
33. Whelan AM, Johansen KL, McCulloch CE, et al. Longer Distance From Dialysis Facility to Transplant Center Is Associated With Lower Access to Kidney Transplantation. Transplant Direct. 2020;6(10):e602. [PubMed: 33134482]
34. Formica RN Jr., Schold JD. The Unintended Consequences of Changes to the Organ Allocation Policy. J Am Soc Nephrol. 2023;34(1):14–16. [PubMed: 36719146]

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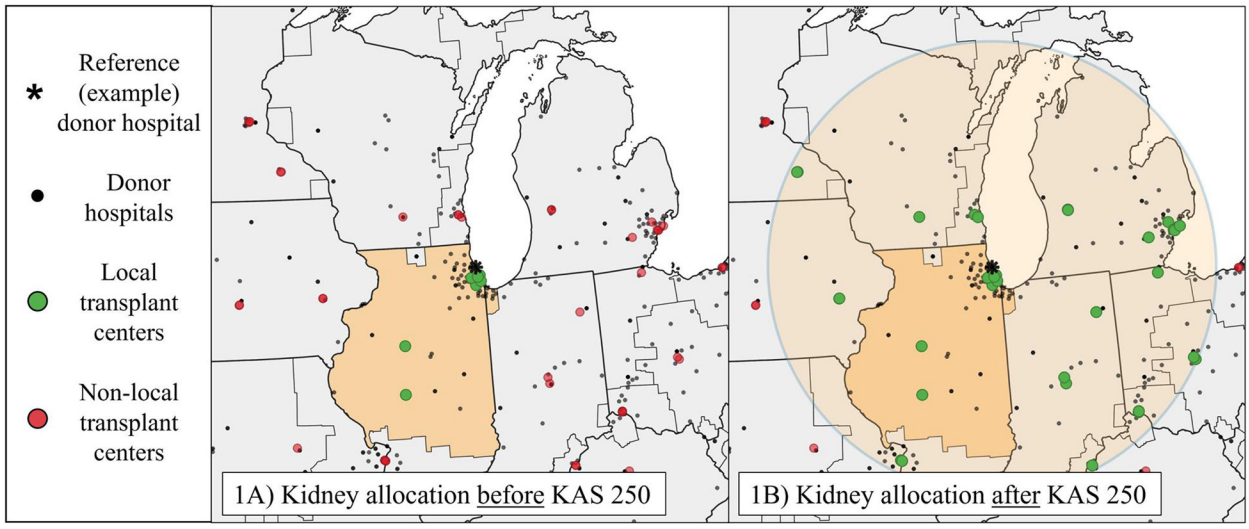

Figure 1A and 1B: Geographic distribution of deceased-donor kidneys under the previous compared to new kidney allocation systems.

KAS 250 = Kidney Allocation System 250 – the newest policy change to broader sharing. The asterisk indicates a reference donor hospital for this example. Before KAS 250 (1A), donor kidneys originating from that hospital would be allocated "locally first" to transplant centers (green circles, N=9) within that hospital's donor service area (yellow shaded area). Red circles indicate non-local transplant centers, who would typically only receive kidney offers in this instance if they had been first declined by all local centers. After KAS 250 (1B), kidneys from the reference donor hospital are now allocated "locally first" to any transplant centers (green circles, N=40) within a 250 nautical mile circle around the reference donor hospital. Prioritization is given to centers closest to the donor hospital within the 250 mile circle.

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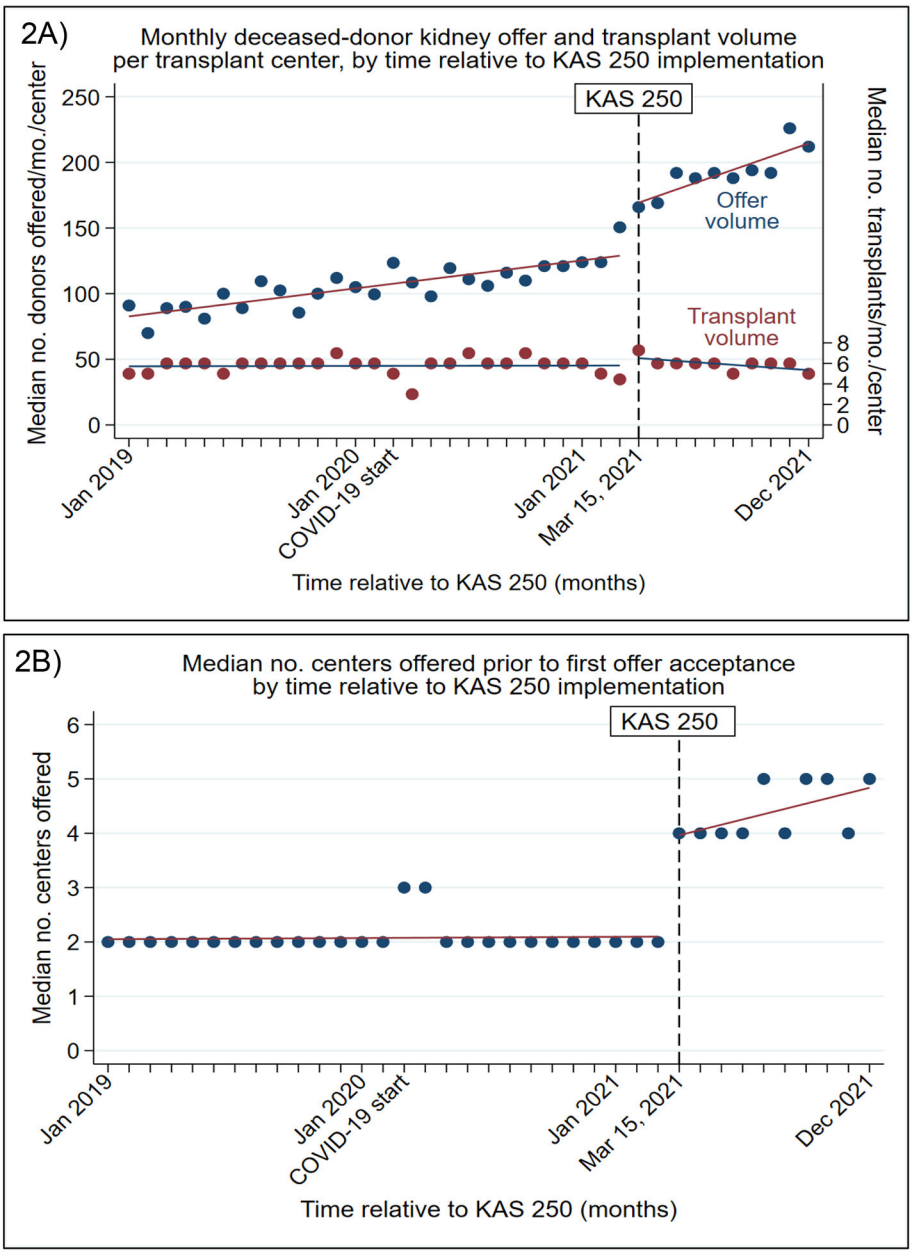
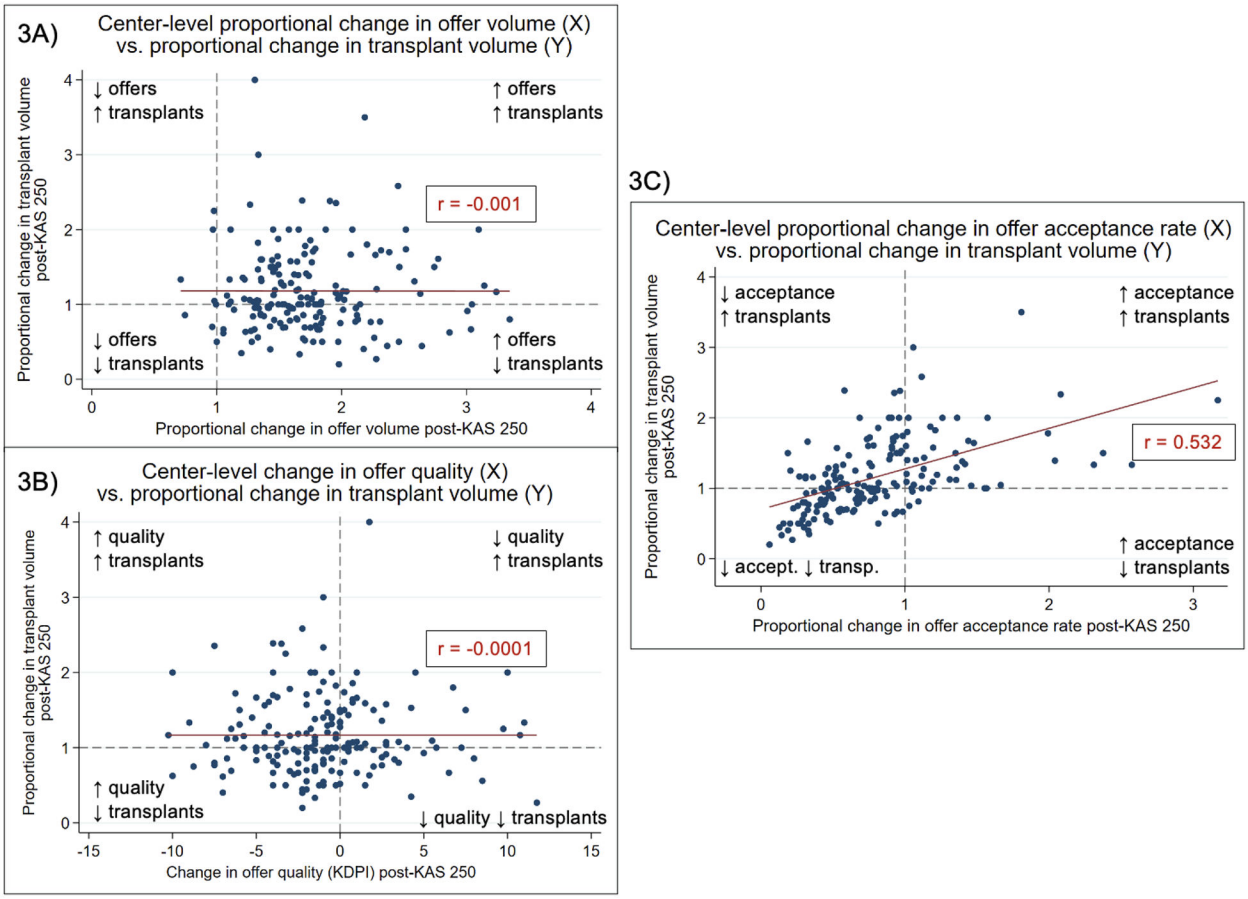


Figures 2A and 2B: Trends in transplant center-specific deceased-donor kidney offer and transplant volume, and efficiency of kidney placement, relative to implementation of kidney allocation policy change.

KAS 250= Kidney Allocation System 250 – the newest policy change to broader sharing. 2A shows the trend over time in the median number of deceased-donor kidney offers received (blue dots, red trend line), and deceased-donor kidney transplants performed (red dots, blue trend line), per transplant center per month. The dashed vertical line indicates the implementation date of KAS 250 (March 15, 2021). 2B shows the trend over time in the median number of transplant centers to whom a deceased-donor is offered before its first kidney is accepted.



Figures 3A-C: Association between center-level changes in deceased-donor kidney transplant volume and kidney offer volume (A), quality of kidneys offered (B), and kidney offer acceptance (C).

KAS 250= Kidney Allocation System 250 – the newest policy change to broader sharing. Each dot indicates one of 185 transplant centers. The y-axis for each plot represents the transplant center-level proportional change in deceased-donor kidney transplant volume in the 9 months after KAS 250 compared to the equivalent 9-month period before KAS 250. The x-axes represent the center-level proportional change in deceased donor kidney offer volume (3A), absolute change in quality of kidney offers received (3B; quality defined by kidney donor profile index, KDPI, with higher numbers indicated lower expected graft longevity), and proportional change in offer acceptance rate (3C) after KAS 250. Outliers were removed from these plots if they were more than 3 standard deviations from the mean (number of centers removed: 2 from 3A, 3 from 3B, 4 from 3C).

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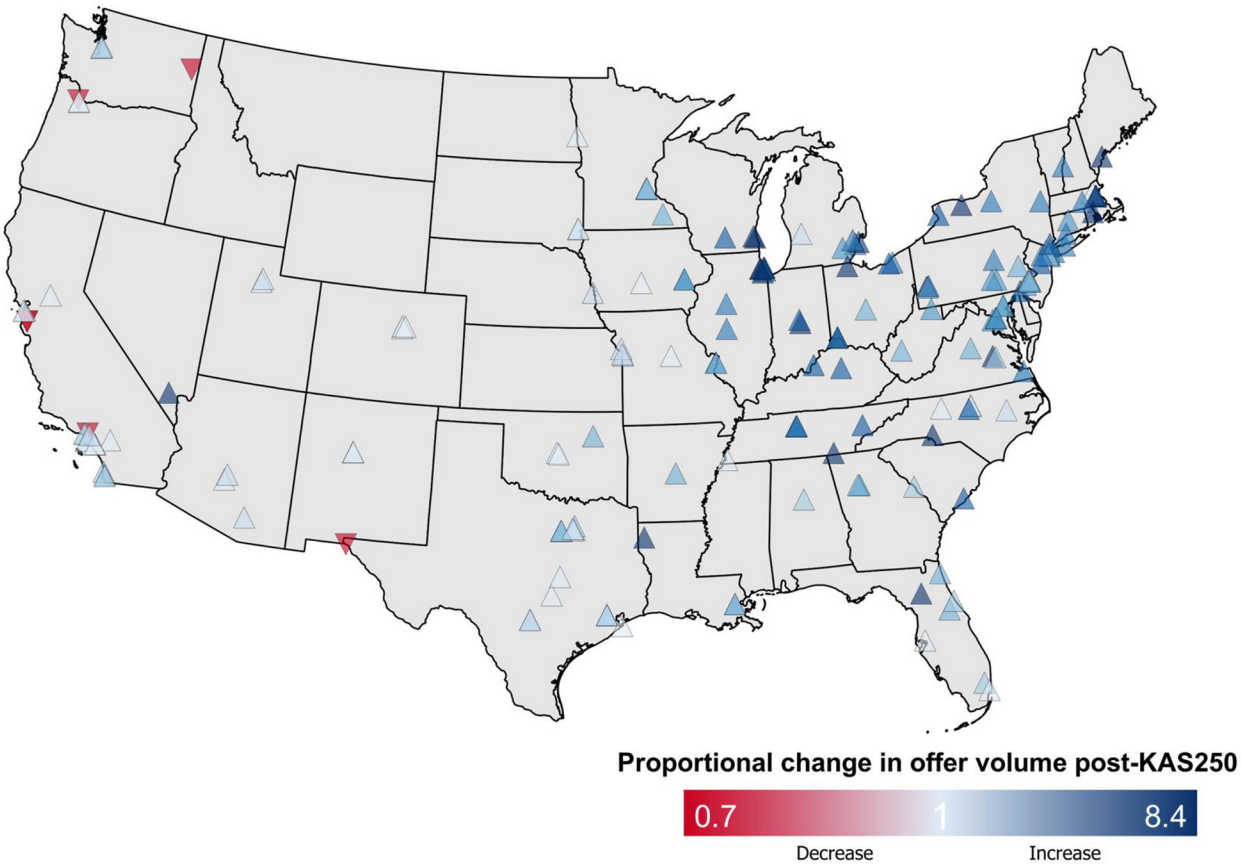

Figure 4A: Location of US kidney transplant centers and their proportional change in deceased-donor kidney offer volume after KAS 250.

Alaska, Puerto Rico, and Hawaii are not depicted here; allocation rules are different in these areas. Each arrow represents a transplant center. Upward facing arrows indicate an increase in offer volume after KAS250, with darker blue indicating a greater increase. Downward facing arrows indicate a decrease in offer volume after KAS250, with darker red indicating a greater decrease.

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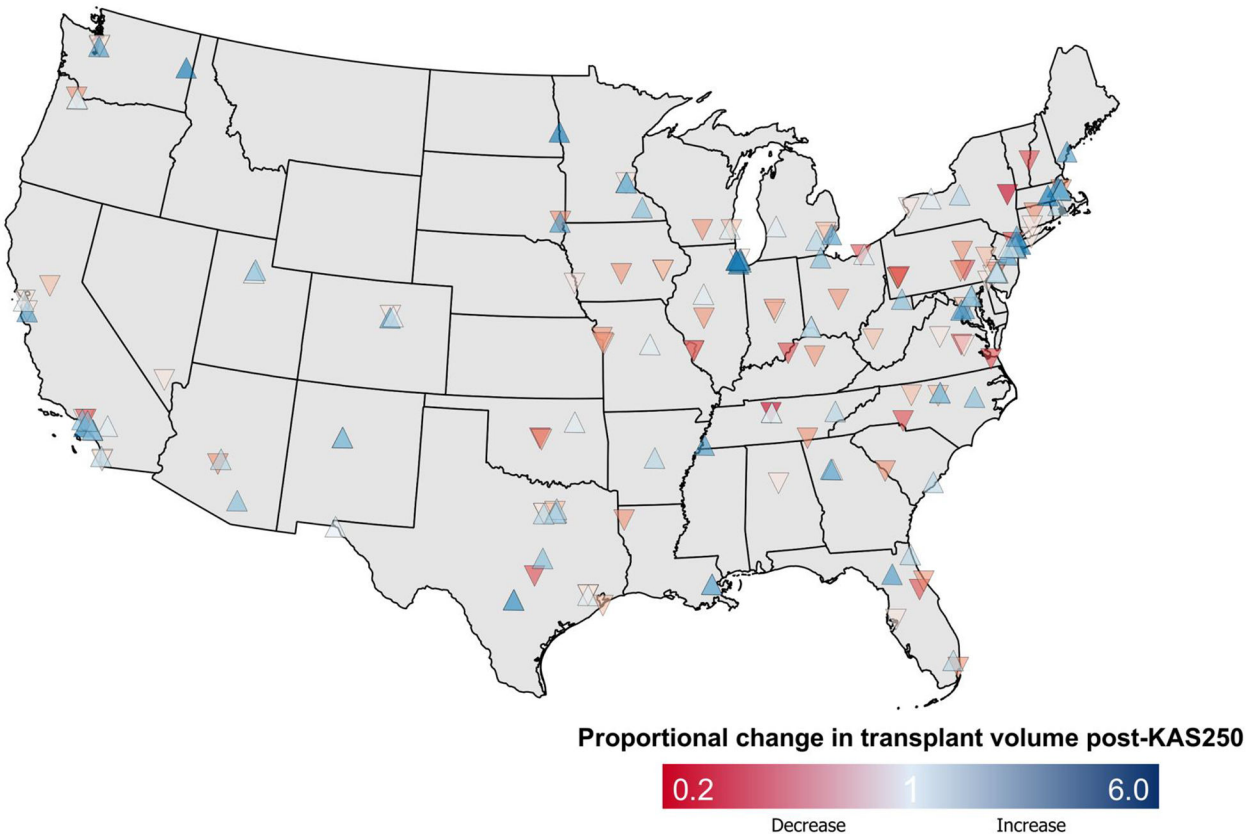

Figure 4B: Location of US kidney transplant centers and their proportional change in deceased-donor kidney transplant volume after KAS 250.

Alaska, Puerto Rico, and Hawaii are not depicted here; allocation rules are different in these areas. Each arrow represents a transplant center. Upward facing arrows indicate an increase in transplant volume after KAS250, with darker blue indicating a greater increase. Downward facing arrows indicate a decrease in transplant volume after KAS250, with darker red indicating a greater decrease.

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Table 1.

Deceased donor-kidney offer and transplant volume, efficiency of placement, and characteristics of donors, before and after implementation of the newest kidney allocation policy.

KAS 250=kidney allocation system 250, the newest allocation policy change which eliminated donor service areas and implemented 250 nautical mile circles to broaden distribution of donor kidneys. KDPI=kidney donor profile index.

Metrics	Pre-KAS 250 (3/15/2020–12/31/2020)	Post-KAS 250 (3/15/2021–12/31/2021)	Proportional increase (post- vs. pre-KAS250)	P-value[d]
Total counts				
Total no. donors	9,395	10,717	1.14	--
Total no. offers	192,152	363,025	1.89	--
National trends				
Total national deceased-donor kidney transplant volume	13,580	14,751	1.09	--
Median cold ischemia time (hrs.)	17.1 (11.5 – 22.7)	19.3 (14.4 – 24.1)	1.13	<0.001
Proportion with delayed graft function	3,844 (28.3%)	4,539 (30.8%)	1.09	<0.001
Proportion of kidneys recovered but not used	3,792 (20.7%)[a]	5,119 (24.4%)[a]	1.18	<0.001
Trends per center per month				
Deceased-donor kidney transplant volume	6 (3 – 10)	6 (3 – 12)	1.00	0.017
No. donors offered, overall (median, IQR)	115 (76 – 151)	195 (137 – 253)	1.70	<0.001
High longevity (KDPI[b] <20)	10 (8 – 12)	17 (13 – 22)	1.70	<0.001
Good longevity (KDPI 21–34)	10 (7 – 13)	18 (13 – 23)	1.80	<0.001
Moderate longevity (KDPI 35–85)	64 (46 – 90)	112 (75 – 145)	1.75	<0.001
Poor longevity (KDPI>85)	26 (14 – 40)	45 (25 – 69)	1.73	<0.001
Efficiency of kidney placement				
No. of centers offered prior to first acceptance, overall (median, IQR)	2 (2 – 2)	4 (4 – 4)	2.00	<0.001
High longevity (KDPI <20)	2 (2 – 2)	3 (3 – 3)	1.50	<0.001
Good longevity (KDPI 21–34)	2 (2 – 2)	4 (4 – 4)	2.00	<0.001
Moderate longevity (KDPI 35–85)	3 (2 – 3)	5 (5 – 5)	1.67	<0.001
Poor longevity (KDPI>85)	5 (4 – 7)	10 (10 – 12)	2.00	<0.001
No. of patients offered[c] prior to first acceptance, overall (median, IQR)	3 (3 – 4)	7 (6–7)	2.33	<0.001
High longevity (KDPI <20)	2 (2 – 2)	5 (4 – 5)	2.50	<0.001
Good longevity (KDPI 21–34)	3 (2 – 3)	6 (5 – 7)	2.00	<0.001
Moderate longevity (KDPI 35–85)	4 (4 – 5)	8 (7 – 9)	2.00	<0.001
Poor longevity (KDPI>85)	22 (16 – 38)	50 (39 – 64)	2.27	<0.001

[a]Total number of recovered kidneys: 18,346 (pre-KAS 250), 20,976 (post-KAS 250).

[b]KDPI is a percentile score ranging from 0 (highest expected graft longevity) to 1 (lowest expected graft longevity). Calculated using each year's corresponding mapping table.

[c]This is the "sequence number" of the match run (how far down the list the kidney was placed).

[d]P-value obtained from Wilcoxon rank-sum or Fisher's exact test (independent samples), or Wilcoxon signed-rank test (dependent samples, i.e. center-specific comparisons).

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Table 2.

Characteristics of transplant centers by proportional change in deceased-donor kidney offer volume, relative to median center, before and after KAS250 implementation.

KAS 250=kidney allocation system 250, the newest allocation policy change which eliminated donor service areas and implemented 250 nautical mile circles to broaden distribution of donor kidneys. The two comparison groups are centers below vs. above the median proportional change in offer volume post-policy change (proportional change <=1.70 vs. >1.70). Numbers represent median (IQR). Waiting time, sensitization, and estimated post-transplant survival all contribute to a patient's priority on the waitlist.

Transplant center characteristic	Decrease or smaller increase in offer volume post-KAS250	Larger increase in offer volume post- KAS250	P-value
Total no. transplant centers	93	92	--
Proportional change in offer volume post-KAS 250 (range)	0.7 – 1.7	1.7 – 8.4	--
Pre-KAS250 deceased-donor kidney transplant volume [a]	73 (34 – 109)	49 (24 – 89)	0.026
Post-KAS250 deceased-donor kidney transplant volume [b]	70 (38 – 125)	48 (26 – 107)	0.043
Pre-KAS250 living donor kidney transplant volume [a]	14 (4 – 27)	13 (6 – 25)	0.963
Post-KAS250 living donor kidney transplant volume [b]	19 (6 – 36)	16 (8 – 35)	0.848
Kidney transplant waitlist population [c]			
Total no. patients	54,656	42,184	
Waitlist size (per center)	370 (217 – 801)	373 (186 – 591)	0.201
Waiting time (years)[d]	2.8 (2.3 – 3.5)	2.7 (2.3 – 3.1)	0.155
Proportion of highly sensitized patients (panel reactive antibody titer >98%)	3.3% (2.3% – 4.5%)	3.1% (2.2% – 4.2%)	0.155
Proportion with high estimated post-transplant survival (score <20)[e]	23.8% (20.1% – 27.5%)	23.7% (21.7% – 27.0%)	0.802

[a]Total volume from March 15, 2020 - December 31, 2020

[b]Total volume from March 15, 2021 – December 31, 2021

[c]Prevalent waitlist population on March 15, 2021

[d]Time since dialysis start or waitlist date (whichever is earlier)

[e]Estimated post-transplant survival is a percentile score ranging from 0 (highest estimated survival) to 100 (lowest estimated survival). This is calculated using the 2021 reference